Formula Cheat Sheet

X stands for X_C or X_L; I_X stands for I_C or I_L. The same formulas work for both circuit types.

Quantity	Formula	Notes
Branch voltages	$E_T = E_R = E_C (or E_L)$	Same voltage on every parallel branch
Resistor current	$I_R = E_T R$	In phase with E
Reactive current	$I_X = E_T X$	Leads (C) or lags (L) E by 90°
Total current	$I_T = (I_R + I_X)$	Vector sum, never I_R + I_X
Impedance	$Z = E_T I_T$	Check: Z = (R × X) ÷ √(R² + X²)
Power factor	$PF = I_R I_T = ZR = WVA$	Decimal between 0 and 1
Phase angle	$Ø = cos(PF) = tan(I_X I_R)$	Leading for RC, lagging for RL
True power	$W = E_T I_R = I_R R$	Resistor only
Reactive power	$VARs = E_T I_X = I_X X$	Reactance only
Apparent power	$VA = E_T I_T = (W + VARs)$	What the source supplies

Solving order used below: voltages → I_R → I_X → I_T → Z → PF → Ø → Watts → VARs → VA. Each answer is then checked a second way (product-over-sum for Z, √(W² + VARs²) for VA).

Metric prefixes: k = ×1,000; M = ×1,000,000; m (milli) = ÷1,000; µ (micro) = ÷1,000,000. Example: 5 V ÷ 10 kΩ = 0.0005 A = 0.5 mA.

Values are rounded to three significant figures; angles to two decimal places.